Lecture 4

Before starting our next topic we will do some mathematical problems related to our previous topic

(For the first 30 min)

❑ Electric Field due to an Electric Dipole

An arrangement with two equal charges but opposite signs is known as an ***Electric Dipole***.

(See Figure 22-8)

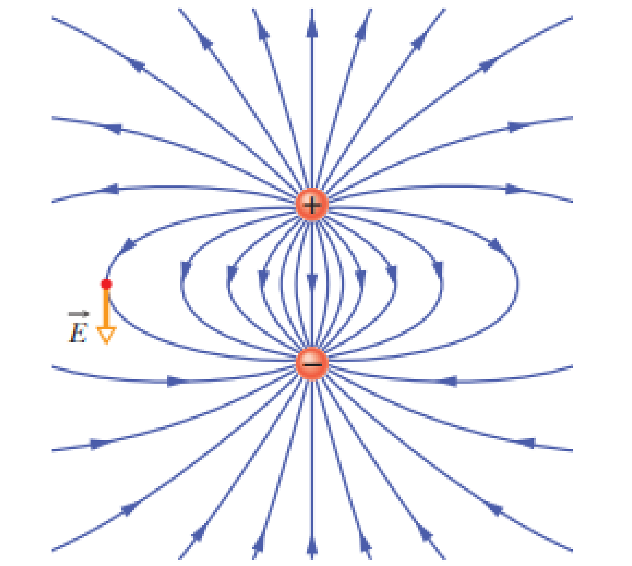
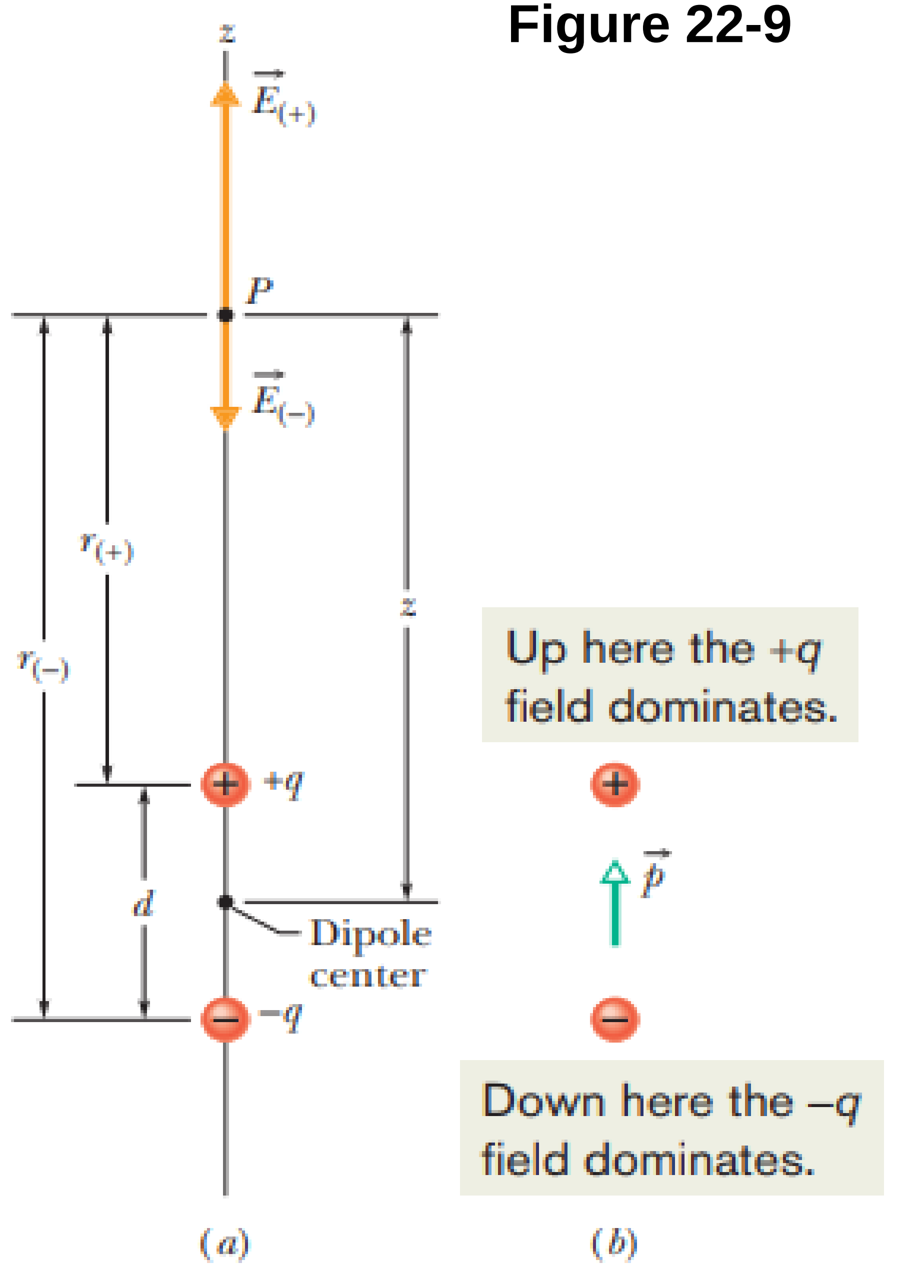


Figure 22-8 The pattern of electric field lines around an electric dipole, with an electric field vector $\vec{E}$ shown at one point (tangent to the field line through that point).

Figure 22-9



Then we can write the magnitude of the net field at ***P*** as

$$
\begin{aligned}
E &= E_{(+)} - E_{(-)} \\
&= \frac{1}{4\pi\varepsilon_0}\frac{q}{r_{(+)}^2} - \frac{1}{4\pi\varepsilon_0}\frac{q}{r_{(-)}^2} \\
&= \frac{q}{4\pi\varepsilon_0(z-\frac{1}{2}d)^2} - \frac{q}{4\pi\varepsilon_0(z+\frac{1}{2}d)^2}.
\end{aligned}
$$

$$E = \frac{q}{4\pi\varepsilon_0 z^2}\left(\frac{1}{\left(1-\frac{d}{2z}\right)^2} - \frac{1}{\left(1+\frac{d}{2z}\right)^2}\right).$$

$$E = \frac{q}{4\pi\varepsilon_0 z^2}\,\frac{2d/z}{\left(1-\left(\frac{d}{2z}\right)^2\right)^2}$$

$$= \frac{q}{2\pi\varepsilon_0 z^3}\,\frac{d}{\left(1-\left(\frac{d}{2z}\right)^2\right)^2}.$$

If $z \gg d$, then $\frac{d}{2z} \ll 1$

$$E = \frac{1}{2\pi\varepsilon_0} \frac{qd}{z^3}.$$

The product qd, which involves the two intrinsic properties q and d of the dipole, is the magnitude $\vec{p}$ of a vector quantity known as the electric dipole moment of the dipole. (The unit of $\vec{p}$ is the coulomb-meter.) Thus, we can write

$$E = \frac{1}{2\pi\varepsilon_0} \frac{p}{z^3}$$

❖ Continuous distribution of charges

1. Line of charge
2. Ring of charge
3. Disk of charge etc.

➢ Charge density

Charge per unit length, area, or volume.

1. Linear charge density, $\lambda = \frac{q}{L}$
2. Surface charge density, $\sigma = \frac{q}{A}$
3. Volume charge density, $\rho = \frac{q}{v}$

The general formula of the electric field for continuous charge density,

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2}$$

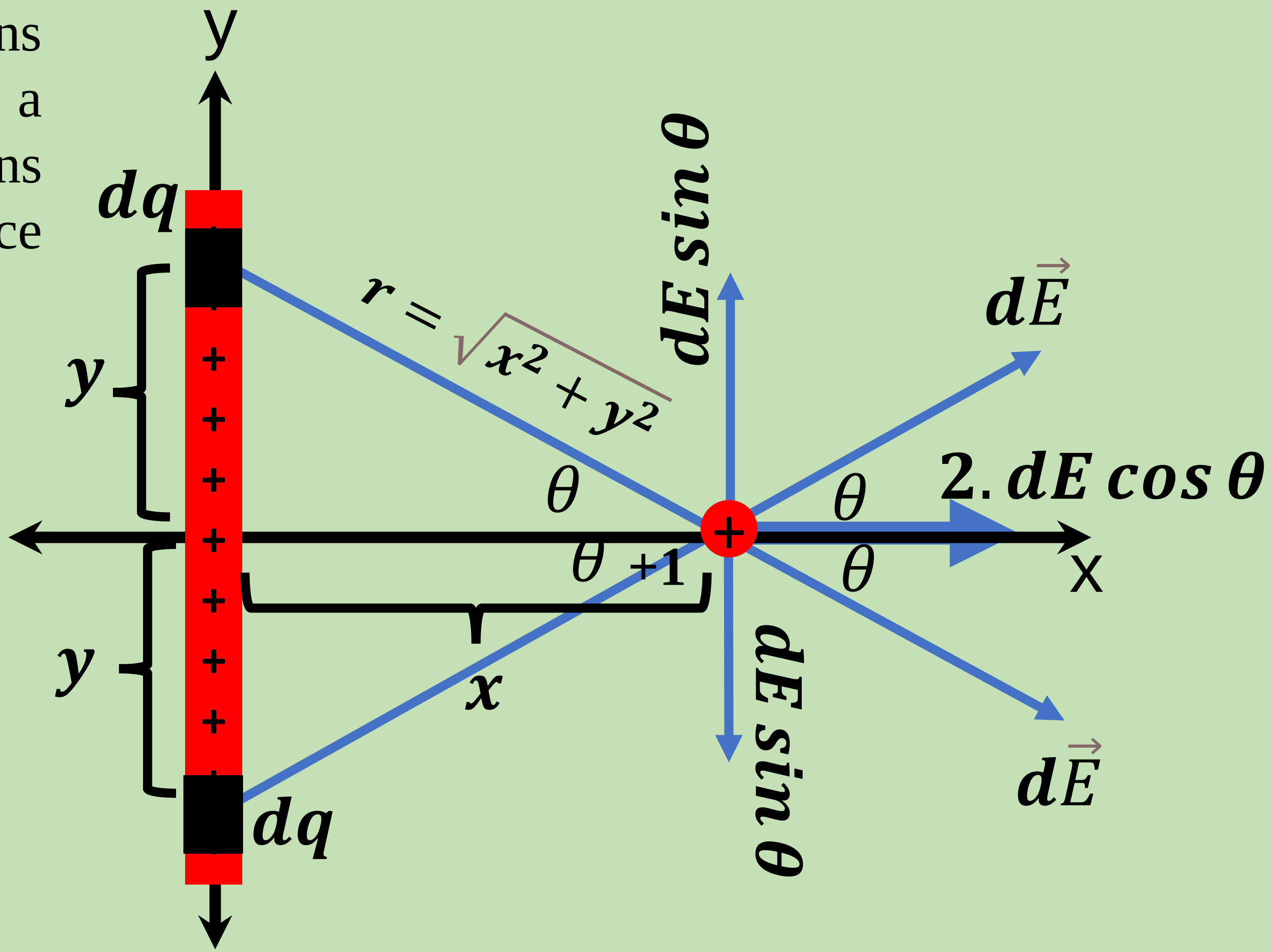
❑ Electric Field Due to a Line of Charge

Let, a rod (charge is uniformly distributed) with a length of L contains $\boldsymbol{q}$ charge with charge density, $\boldsymbol{\lambda}$ and a small segment $\boldsymbol{dy}$ on its body contains the charge of $\boldsymbol{dq}$ situated at a distance $\boldsymbol{y}$ from the origin.

The electric field, $\boldsymbol{dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2}}$

Charge density, $\boldsymbol{\lambda = \frac{q}{L} = \frac{dq}{dy}}$

Therefore, $\boldsymbol{dE = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, dy}{r^2}}$



$$\cos\theta = \frac{x}{r} = \frac{x}{\sqrt{x^2 + y^2}}$$

$$E = \int_0^{+L/2} 2.\, dE\cos\theta$$

$$= \int_0^{+L/2} \frac{2}{4\pi\varepsilon_0} \frac{\lambda\, dy}{(x^2+y^2)} \frac{x}{\sqrt{x^2 + y^2}}$$

$$= \frac{2\lambda x}{4\pi\varepsilon_0} \int_0^{+L/2} \frac{dy}{(y^2+x^2)^{\frac{3}{2}}}$$

$$E = \frac{1}{4\pi\varepsilon_0} \frac{\lambda L}{x\sqrt{x^2 + \frac{L^2}{4}}}$$

$$\int \frac{dx}{(x^2 \pm a^2)^{\frac{3}{2}}}$$

$$= \frac{\pm x}{a^2\sqrt{(x^2 \pm a^2)}}$$

In the limit, $x \to \infty$, the equation approaches to the equation of the electric field of a point charge.

$$E = \frac{1}{4\pi\varepsilon_0} \frac{q}{x^2};\ \lambda = qL$$

Again, if $L \gg x$ *(infinitely long line of charge)*, then the electric field will be

$$E = \frac{\lambda}{2\pi\varepsilon_0 x}$$

•••32 GO In Fig. 22-55, positive charge $q = 7.81\ \mathrm{pC}$ is spread uniformly along a thin nonconducting rod of length $L = 14.5$ cm. What are the (a) magnitude and (b) direction (relative to the positive direction of the x axis) of the electric field produced at point P, at distance $R = 6.00$ cm from the rod along its perpendicular bisector?

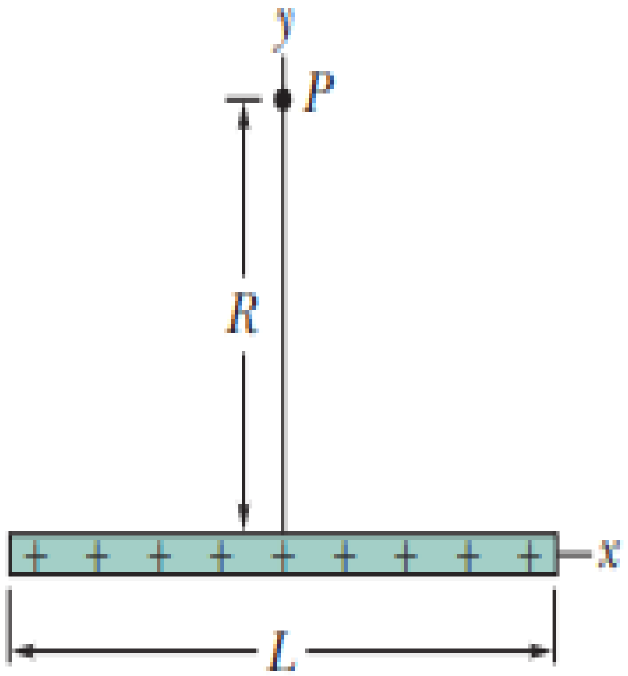


Figure 22-55 Problem 32.

❑ Electric Field Due to a Ring of Charge

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, ds}{r^2}.$$

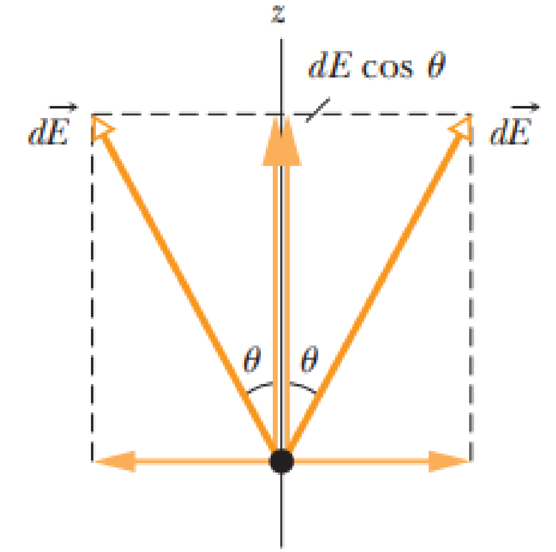
$$dE = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\, ds}{(z^2 + R^2)}.$$

$$dE\cos\theta = \frac{1}{4\pi\varepsilon_0}\frac{z\lambda}{(z^2 + R^2)^{3/2}}\,ds.$$

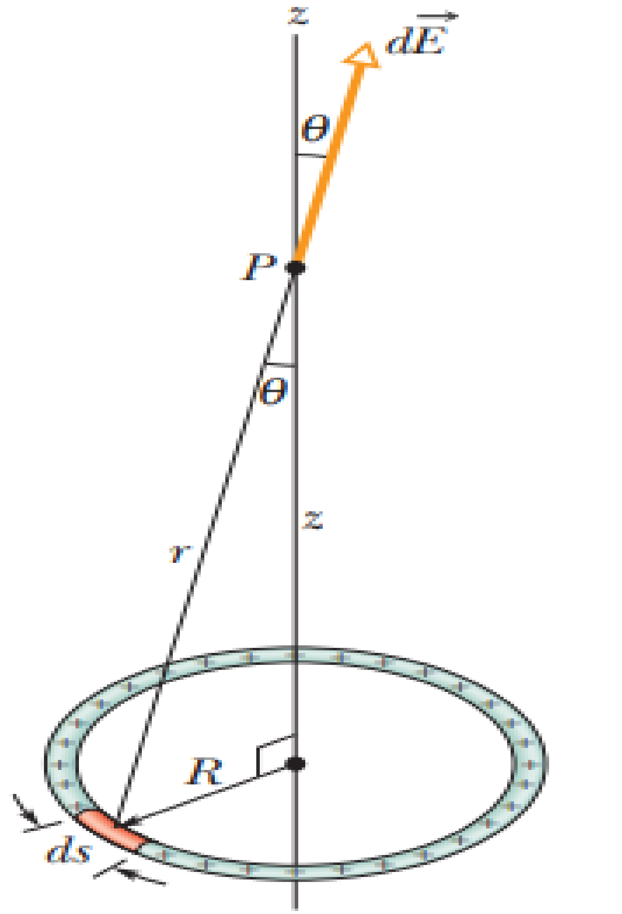
$$E = \int dE\cos\theta = \frac{z\lambda}{4\pi\varepsilon_0(z^2 + R^2)^{3/2}}\int_0^{2\pi R} ds$$

$$= \frac{z\lambda(2\pi R)}{4\pi\varepsilon_0(z^2 + R^2)^{3/2}}.$$

$$E = \frac{1}{4\pi\varepsilon_0}\frac{qz}{(z^2+R^2)^{\frac{3}{2}}};\ where,\ \lambda = \frac{q}{2\pi R}$$



$$\cos\theta = \frac{z}{r} = \frac{z}{(z^2 + R^2)^{1/2}}.$$



If $z << R$, then $z^2 + R^2 \approx z^2$

Therefore, $E = \frac{1}{4\pi\varepsilon_0}\frac{q}{z^2}$

(Charged ring at large distance looks like point charge)

If an electron is constrained to the central axis of a positively charged ring, what happens for z<<R?

Assume that an electron of charge, ***e*** just release from a point, ***P***

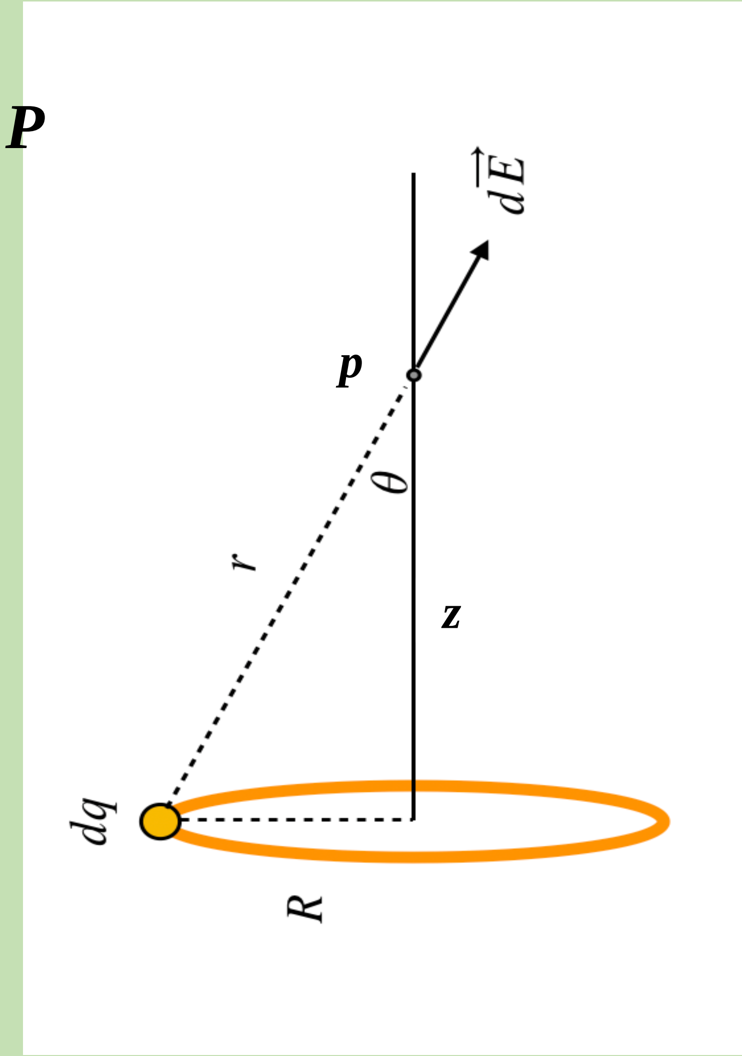
Electric field at point p directed downward along the z-axis for the positively charged ring is given by the following formula,

$$\boldsymbol{E} = \frac{qz}{4\pi\varepsilon_0(z^2+R^2)^{\frac{3}{2}}}$$

Therefore, the force on the electron due to the electric field is

$$\boldsymbol{F} = e\boldsymbol{E}$$

So, $\boldsymbol{F} = \dfrac{qez}{4\pi\varepsilon_0(z^2+R^2)^{\frac{3}{2}}}$



For the case, z<<R

we can write, $\boldsymbol{F} = \dfrac{\boldsymbol{qez}}{\boldsymbol{4\pi\varepsilon_0 R^3}}$

Where, the direction of force, F and z is opposite each other

Therefore, $\boldsymbol{F} = -\boldsymbol{kz}$

Where, $\boldsymbol{k} = \dfrac{\boldsymbol{qe}}{\boldsymbol{4\pi\varepsilon_0 R^3}}$ (constant)

So, the electron is in the SHM

••24 A thin nonconducting rod with a uniform distribution of positive charge Q is bent into a complete circle of radius R (Fig. 22-48). The central perpendicular axis through the ring is a z axis, with the origin at the center of the ring. What is the magnitude of the electric field due to the rod at (a) $z = 0$ and (b) $z = \infty$? (c) In terms of R, at what positive value of z is that magnitude maximum? (d) If $R = 2.00$ cm and $Q = 4.00\ \mu C$, what is the maximum magnitude?

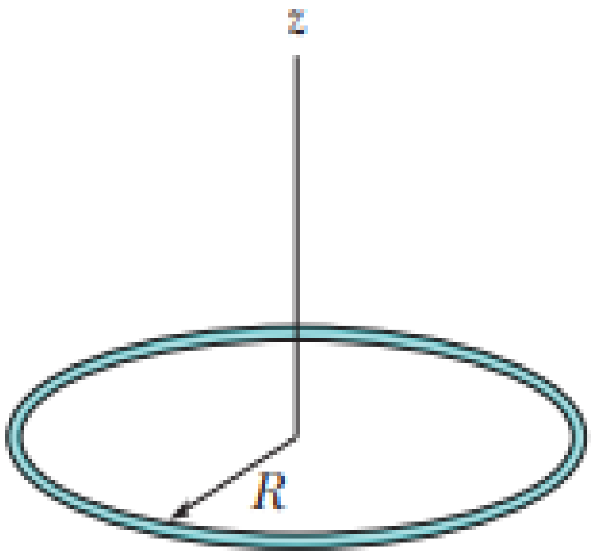


Figure 22-48 Problem 24.

Thank You